In 2020, expect web 4.0
The numbers next to the web like web 2.0 and 3.0 roughly correspond to the decade of the existence of the internet.

virtual worlds such as *Second Life*. Scientists will be able to use these basic frameworks to design their own visions of the internet, and then test those designs using GENI.

On the other hand, there's the PlanetLab project, which aims to create *overlay* technologies, which will sit on top of the world's existing internet infrastructure, and by becoming more popular, will squish out the old technology. Something like this has already been in existence, in the form of the university-only Internet2. The underlying philosophy is that there are several parts of the internet that get the job done just fine, so why throw all that away?

So what does a more realistic internet look like?

Middle ground

While we wait for the competing projects to come up with a brand new internet we can use and love, there are two upcoming technologies that we should see pretty soon. You already know about the first — IPv6. The other is called *flow routing*.

Right now, packets on the internet travel like they always have — taking different paths to their destination, where they are reconstructed. In a flow-based network, packets that need to travel together — voice, audio and video, for example — will travel together. The new flow routers will work like regular routers, except when they see packets that say that they're part of a *flow*. For these packets, the routers will choose paths that allow the packets to reach their destination in order.

Of course, a flow router won't be able to do its work without packets that have information flows. And this is where IPv6 comes in. IPv6 packets have *traffic class fields*, which will tell routers how to prioritise them. As a bonus, these packets will also carry *authentication* and *security* headers, which will let routers and PCs find out whether the packets are coming from the right server.

Finally, IPv6 opens up a whole new world of IP addresses, which means that every device — from PC to phone to hearing aid — can have its own unique address, and not have to hide behind access points ever again. This means it's theoretically possible to have an IP address that never changes. So, if you're using a service that authenticates you based on your IP address, you don't have to worry about losing your authorisation by moving to a new network.

However, even though IPv6 was standardised in 1998, only 0.236 per cent of all internet users are using IPv6, according to a Google report published in October 2008 (read the PDF at **http://tinyurl.com/6c634k**). It may be disappointing news now, but it does offer a simple solution to our IPv6 adoption problem. The people on IPv6 may not necessarily be using it intentionally — their ISPs began to support IPv6, so they gave customers IPv6-capable routers. And since the newest generations of Windows, Mac OS X and Linux are all set to prefer IPv6 if it's available, more people went online with the new protocol without even realising it. Which means that it isn't the people who need to be convinced, it's the ISPs.

So after all this talk, it may well be that the future of the internet is simple — slightly smarter, with a new protocol and updated routers.

Which may be a good thing, because we may not want a smart internet after all...

The stupidity of smartness

We point nasty fingers at dumb networks, but the truth is that even smart networks aren't beyond reproach. When you load routers with software that authenticates and analyses and directs and decides, you're actually getting in the way of data moving swiftly. But there's a possibility that's even worse. We complain about the internet today, because it's an old design that hasn't been improved because it got too large — but twenty years in the future, people will say the same thing about a smart network. Take the telephone network, for instance — it's the original smart network, fully aware of the signals that travel along it — it has now solidified into a behemoth that's too expensive to upgrade, and too complicated to easily add new functionality.

Tom Evslin, who helped build the original MSN, and AT&T's first internet service, thinks that any effort to make the internet smarter "will cost incalculably more in loss of future flexibility and scalability." He argues that the dumbness of the internet is what enables it to support new and wonderful applications, and should thus stay that way. He isn't the only one to think this way, either. Vinton Cerf, Google's Chief internet Evangelist, and *father of the internet*, says that it's the terminals that need to get smarter, not the network.

And then, there are the "political" implications. A fully redesigned internet may well turn out to be what the *New York Times* calls a "gated community where users would give up their anonymity and certain freedoms in return for safety." Scary thought, yes?

And when you think about it, a lot of news about internet outages seems overblown. In October, a worm called Conficker infected 15 million Windows PCs, prompting renewed fears of a "digital Pearl Harbour", which will supposedly bring the internet to a standstill. But if Conficker was the harbinger of doom, why didn't you stop surfing? Did you even notice that the internet "was under attack"? Truth is, Conficker didn't bring down the internet, it exploited a vulnerability in an OS. A *Windows* OS. If there's any terminal that needs to get smarter about security, it's the Windows terminal.

Bottom line: whatever the internet ends up being, we shouldn't accept a design that emerged from panic. ■

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